

MadhuTwin

A hybrid digital twin for Type 2 Diabetes that fuses EHR and wearable data to warn doctors up to 2 hours before glucose spikes and lows

MODEL

SYNC

SIMULATE

Team ClockItTwin · IIT Kharagpur

Happiest Health Digital Twin Challenge 2026 · Phase 1

Diabetes care in India is reactive

101 million

adults in India live with diabetes

136 million

more have prediabetes

Source: ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023

A quarterly HbA1c describes the past

It cannot warn about tonight's low on a sulfonylurea or premixed insulin, or tomorrow's illness-driven rise.

Post-meal spikes go unseen

Rice- and roti-heavy meals, late dinners and festival sweets drive excursions that clinic visits never capture.

Data exists, attention does not

CGMs and smartwatches stream thousands of readings a day; no care team can watch hundreds of patients' streams.

MadhuTwin: a living virtual replica of each patient

Not just a predictor: the three defining properties of a digital twin

MODEL

Personalised glucose–insulin physiology

An extended minimal model with meals, exercise, sleep, dawn effect and Indian therapies, fitted to each patient with EHR-based priors.

SYNC

Continuously updated from live data

An Unscented Kalman Filter re-estimates hidden insulin state and a drifting insulin sensitivity every 5 minutes.

SIMULATE

Run forward to predict and explore

Causal 2-hour forecasts and what-ifs: a different meal, a walk after dinner, a short night, an illness.

On top: TwinNet, a physics-gated network that fuses both data streams with the twin's own forecast. It is then fine-tuned to each person and explains every alert in plain language.

Architecture: two data streams fused into the twin

MadhuTwin architecture: two data streams fused into a living digital twin

Type 2 Diabetes · predicts glucose spikes and hypoglycaemia up to 2 hours ahead · doctor-facing dashboard

1 · Data streams

Static / historical: EHR

Demographics, diagnoses (SNOMED CT)
Labs: HbA1c, FPG, insulin, lipids, eGFR (LOINC)
Medications with timing (WHO ATC)
Family history · TCF7L2 rs7903146 · polygenic risk
FHIR R4 bundles, ABDM-aligned

Dynamic / real-time: wearables & IoT

CGM every 5–15 min
Heart rate, HRV (RMSSD), steps, METs
Sleep stages (hypnogram)
Logged meals (Indian foods) and doses

Sources

Synthetic India cohort 1,000 × 14 d · CGMacros (45, real) · BIG IDEAs (16, real HRV) · ShanghaiT2DM (100, external)

2 · Digital twin core

Ingest & fuse

Common 5-min schema · missing-aware · EHR vector + 21 dynamic channels

MODEL

Personalised glucose–insulin physiology (MAP fit, EHR priors)

SYNC

Unscented Kalman filter every 5 min: state + insulin-sensitivity drift

SIMULATE

2-h forecasts and what-if: meals, walks, sleep, illness

TwinNet: physics-gated fusion network

Causal dilated TCN over 6 h × 21 channels
EHR encoder conditions the dynamics (FiLM)
Gate learns how much to trust the physics twin per horizon
Outputs: 5–120 min quantiles (conformal 80% band) + spike / hypo probability
Fine-tuned per person, averaged with LightGBM; SHAP reasons

3 · Clinician experience

FastAPI + WebSocket

Live replay · audit trail · FHIR export

Doctor dashboard

Risk-ranked remote-monitoring panel
Virtual patient: organ-level physiology
24 h CGM + 2 h forecast cone + alerts with reasons
AGP report (international consensus)
What-if simulator with Indian meals
Ask the twin (grounded assistant)

Interoperability

FHIR R4 RiskAssessment carries the twin's forecast to any EHR

Governance consent artefacts · purpose-tagged audit log · de-identified & synthetic data · model card · DPDP Act 2023 · CDSCO SaMD pathway

Two data streams, four open or synthetic sources

Sandbox rules respected: no real Indian patient data is used

Static / historical: EHR

Demographics · diagnoses (SNOMED CT) · labs incl. HbA1c, fasting glucose & insulin, lipids, eGFR (LOINC) · medications with timing (WHO ATC) · 2-year lab history · family history · TCF7L2 rs7903146 genotype · polygenic risk score

Dynamic / real-time: wearables & IoT

CGM (5- or 15-min) · heart rate · HRV (RMSSD) · steps & METs · sleep stages · logged meals from an Indian food table · insulin and tablet doses

1,000 × 14 days

45 people

16 people

100 patients

Synthetic India-calibrated patients

ICMR-INDIAB mix, regional diets, Indian prescribing; streams generated by the same physiology

CGMacros (PhysioNet)

Real CGM, Fitbit, meals, labs; healthy to T2D (USA)

BIG IDEAs (PhysioNet)

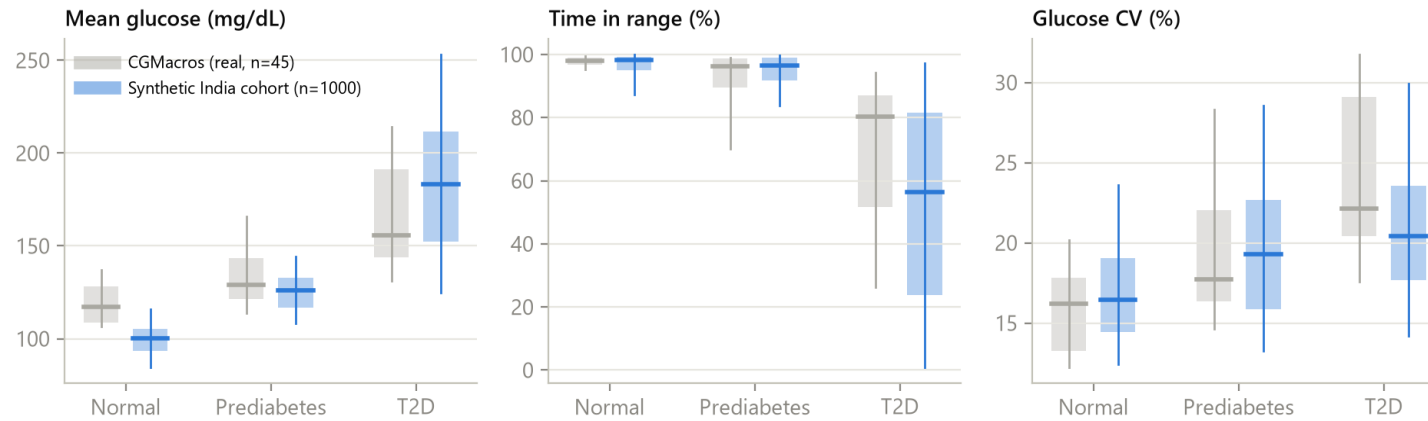
Real wristband heart rate and HRV, Dexcom CGM, food logs (USA)

ShanghaiT2DM (Figshare)

External: rice-based diet, insulin users, no wearables

Synthetic Indian cohort, causally linked streams

Synthetic CGM statistics vs real people (5th-95th, IQR, median); the Indian T2D cohort is deliberately sicker (median HbA1c 8.0% vs 7.1%)



Physiology parameters are sampled from relationships measured by fitting twins to 45 real people, so synthetic glucose behaves like real glucose.

- Status mix, South-Asian BMI phenotype, comorbidities and prescribing calibrated to Indian clinics: premixed 30/70 insulin, gliclazide, teneligliptin
- Regional meals from IFCT 2017: idli-sambar, roti-sabzi, ragi mudde, biryani, chai with sugar, festival sweets
- Wearables are simulated by the same physiology, and labs are measured from the simulated body (HbA1c from mean glucose)
- Realistic imperfection: unlogged meals, carb misestimation, CGM noise and compression lows, missed doses, illness and stress days

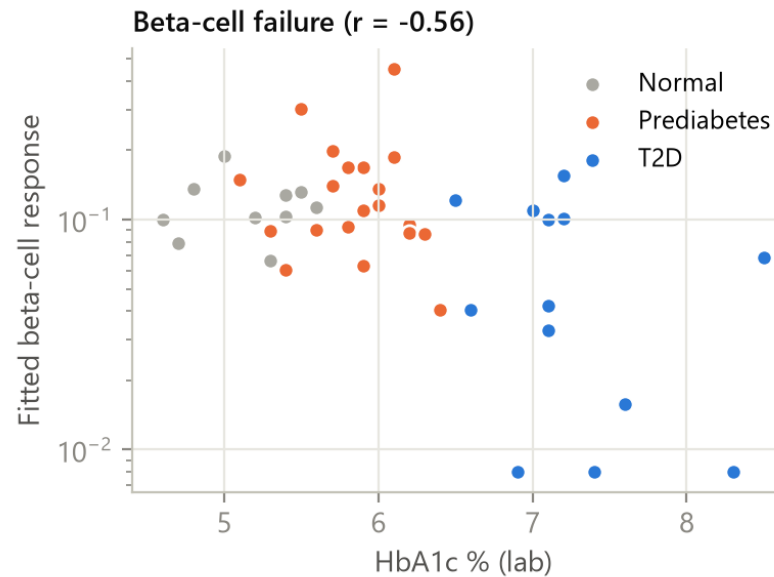
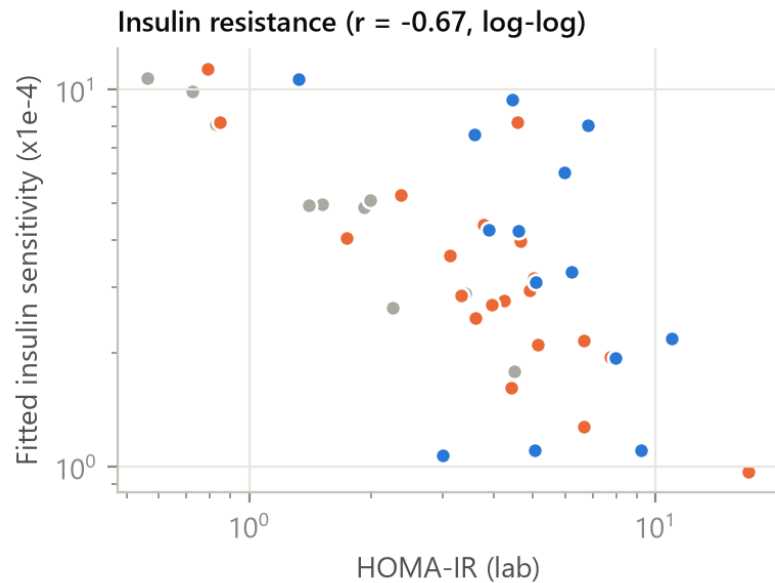
MODEL: twins recover real physiology from CGM

MODEL

SYNC

SIM

Twins fitted to 45 real people recover known physiology from CGM



37.7 → 26.6

mg/dL twin error, EHR prior → personalised (45 real people, 1–4 h windows)

29% lower

error once the twin is fitted to the individual

Fitted insulin sensitivity tracks lab HOMA-IR and beta-cell response tracks HbA1c, without the model ever seeing those relationships during fitting.

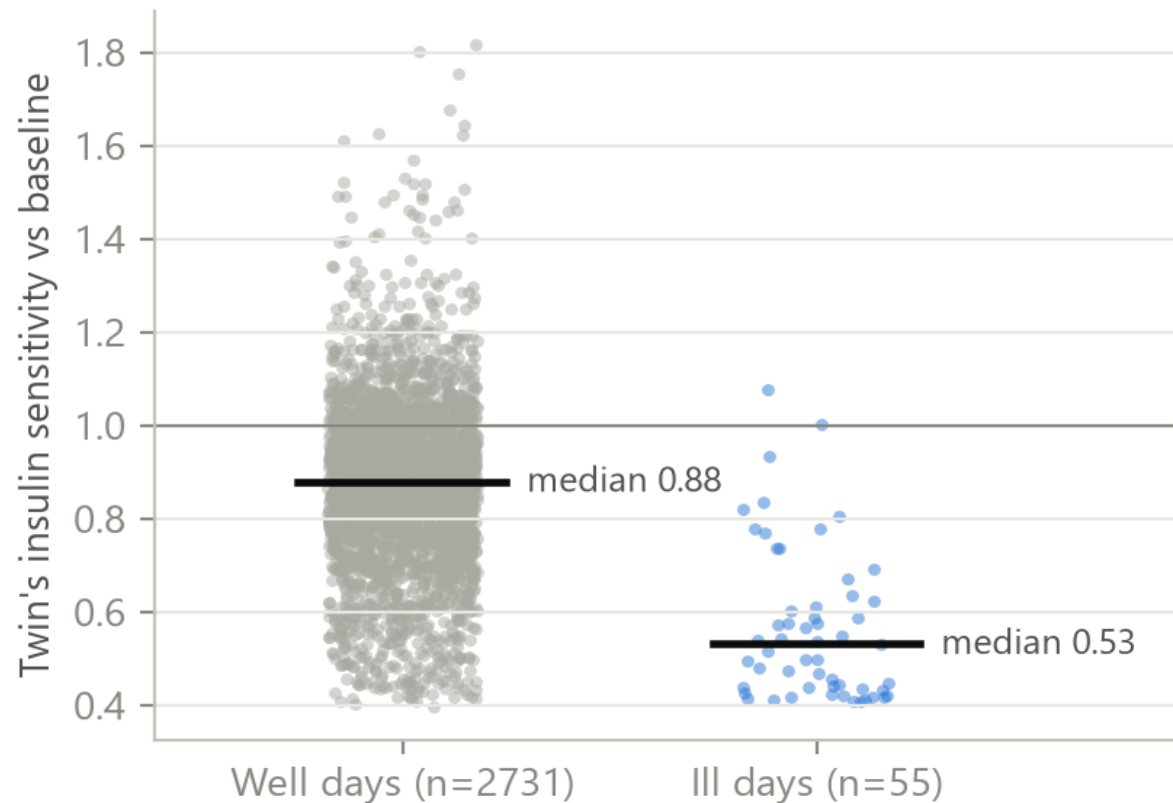
SYNC: the twin notices illness from CGM alone

MODEL

SYNC

SIM

The synced twin detects illness from CGM alone



0.89

AUROC for identifying ill days from the synced insulin-sensitivity trace (55 ill days, 2786 patient-days)

0.57x vs 0.87x

insulin sensitivity relative to baseline on ill vs well days

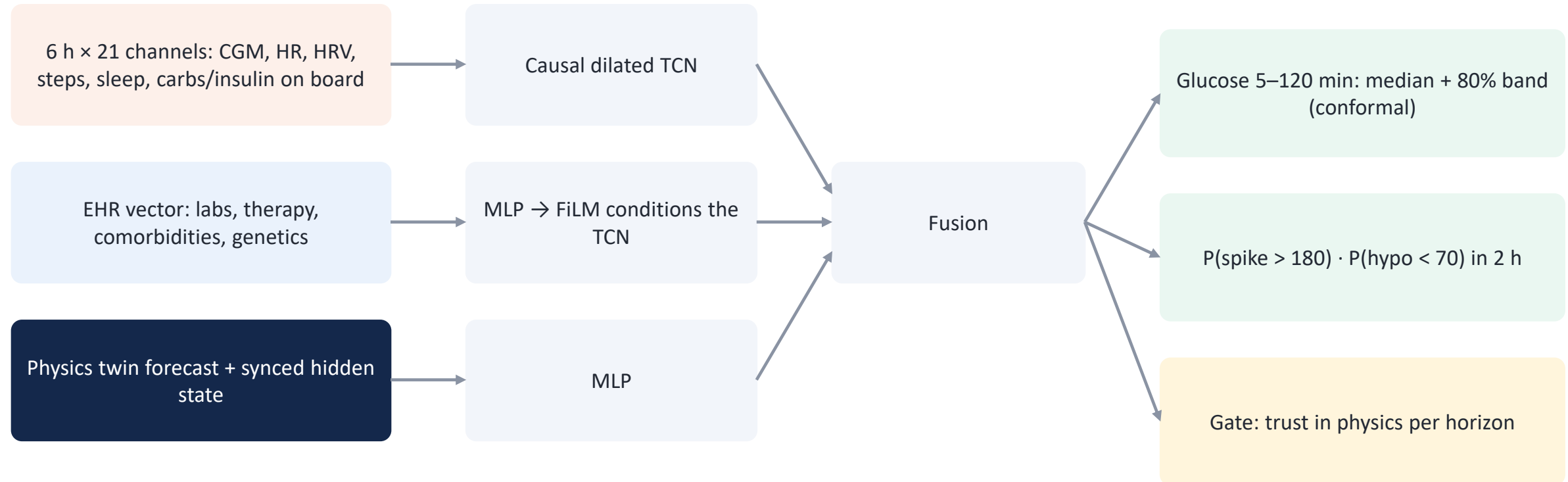
The Unscented Kalman Filter tracks a drifting insulin-sensitivity factor every 5 minutes. A sustained drop prompts the dashboard insight: check for infection, stress or steroid use.

SIMULATE: TwinNet fuses both streams with the twin

MODEL

SYNC

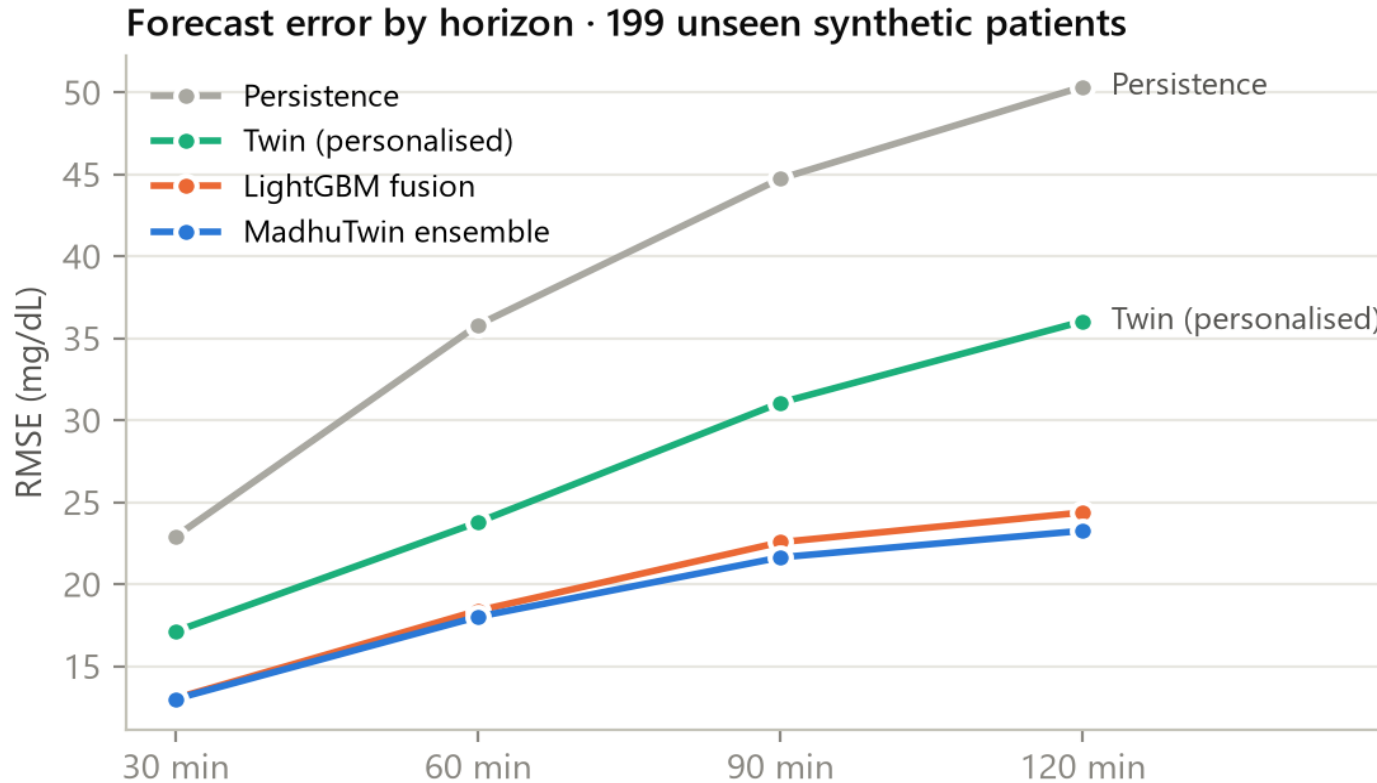
SIM



Learned trust in the physics twin rises with horizon: 0.71 at 30 min, 0.82 at 60 min, 0.92 at 2 h. Modality dropout lets the same model run without wearables or EHR. Pretrained on synthetic patients, fine-tuned on real ones, then on each person: averaged with LightGBM, this is the MadhuTwin ensemble.

Accurate 2-hour glucose forecasts on unseen patients

199 held-out patients · days 8–14 after personalisation · 1,29,956 forecasts



18.0 mg/dL

60-min error (RMSE), MadhuTwin; persistence 35.8 ·
TwinNet 19.0 · LightGBM 18.4

99.5%

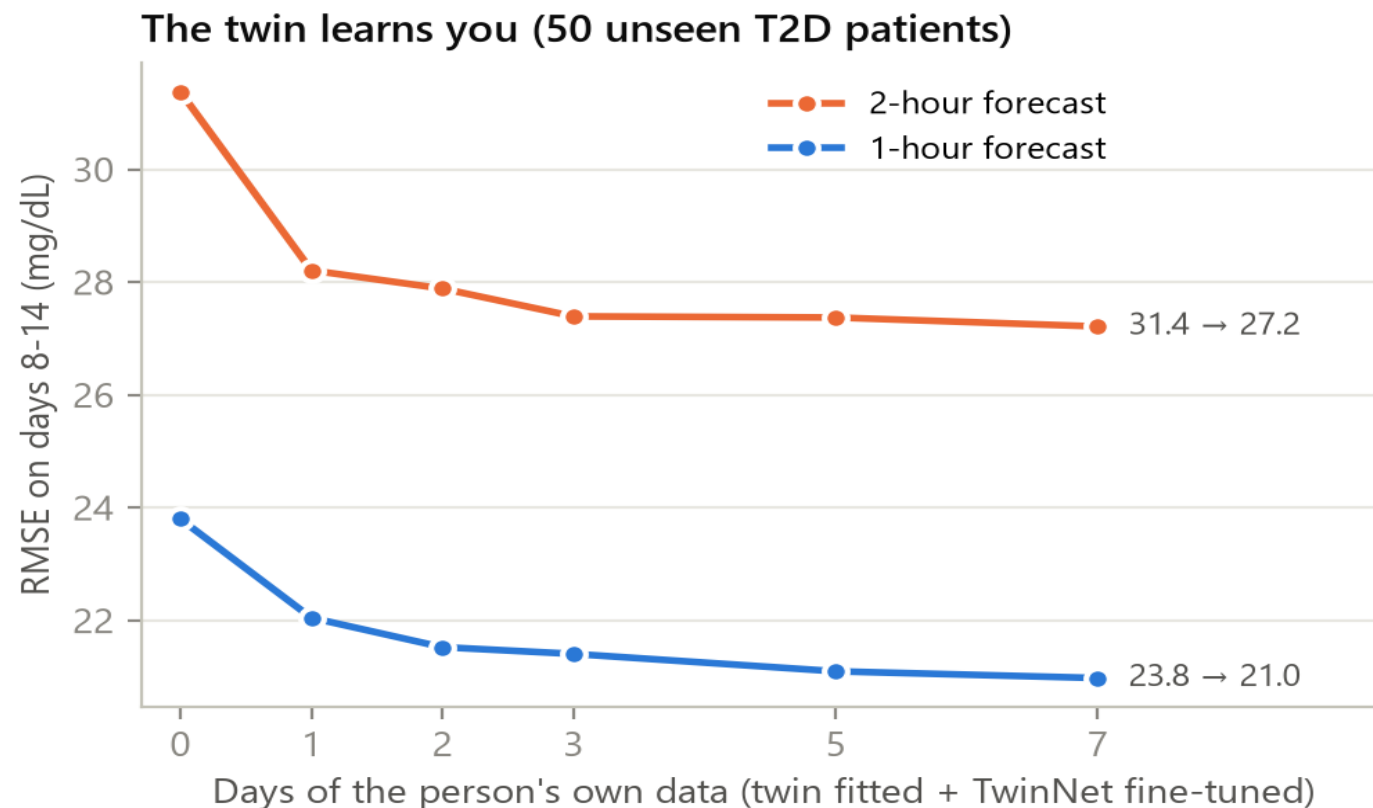
of 60-min forecasts in Clarke zones A+B (clinically
acceptable)

79%

of outcomes inside the 80% interval (target 80%)

The twin learns you

50 unseen people with T2D · twin fitted and TwinNet fine-tuned on their first days · scored on days 8–14



31.4 → 27.2

mg/dL 2-hour error: no personal data → after 7 days of personal data

13% lower

2-hour error after a week (after one day: 28.2 mg/dL)

Fitting the twin and fine-tuning TwinNet take seconds on a CPU, use only the person's earlier data, and can run every night as new CGM arrives.

Early warnings with lead time a clinician can act on

90%

of sustained spikes (>180) flagged in advance

110 min

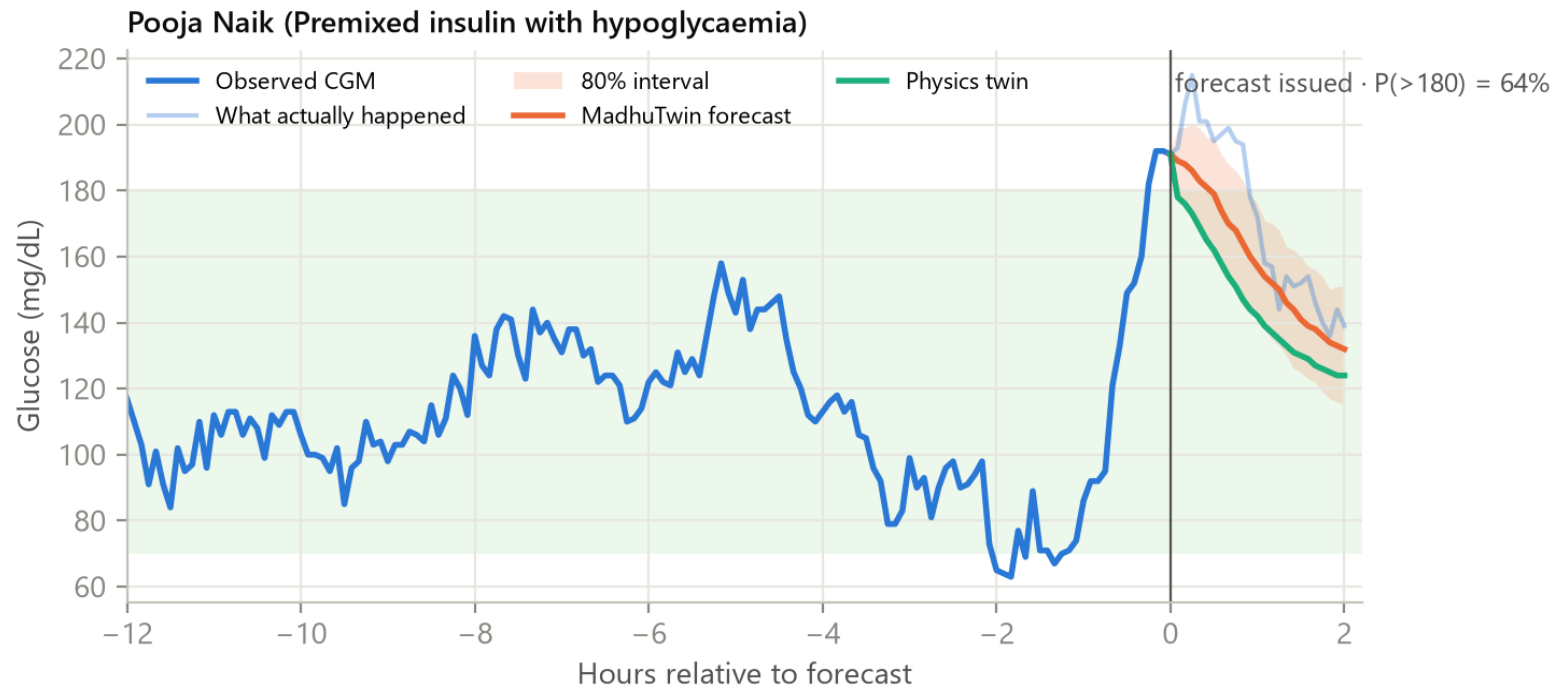
median warning before the spike begins

≤ 1

false alert episodes per patient-day

0.97

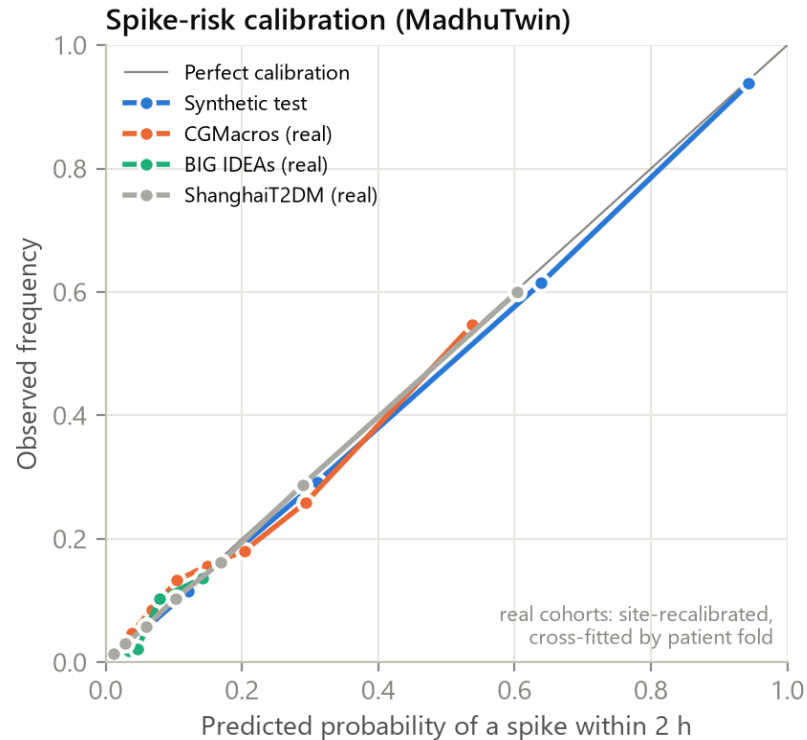
hypoglycaemia AUROC (spike AUROC 0.96)



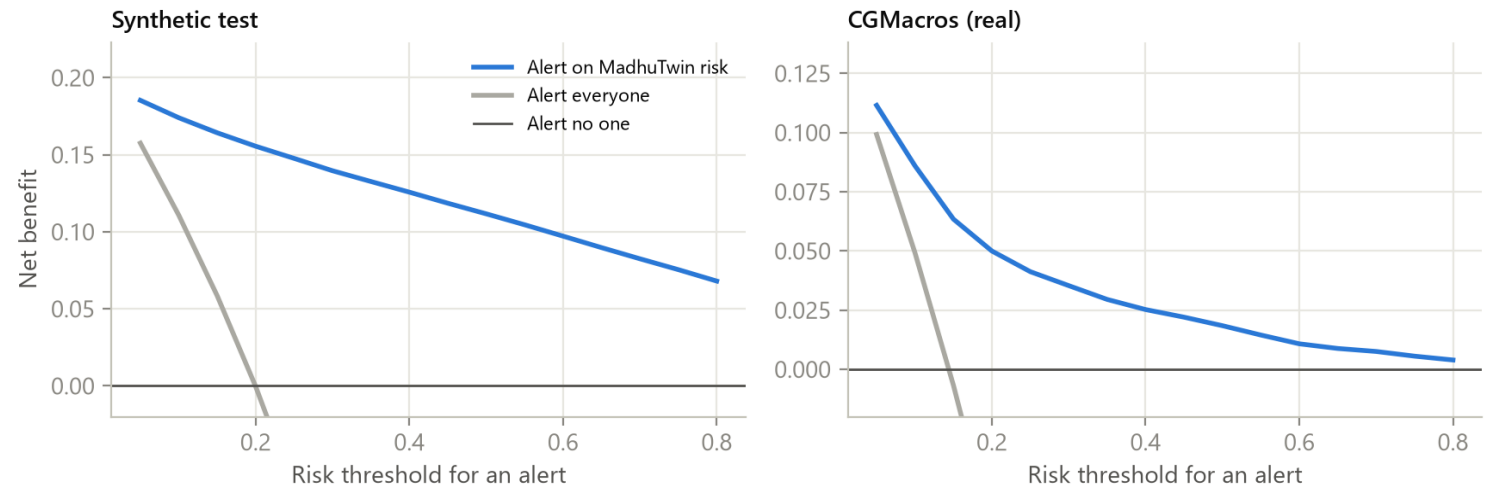
Every alert carries its reasons from the event model, e.g. carbohydrate still absorbing, a rising trend, insulin sensitivity below baseline, or a sedentary hour after a meal.

Calibrated, useful and fair

What a diabetologist asks beyond AUROC

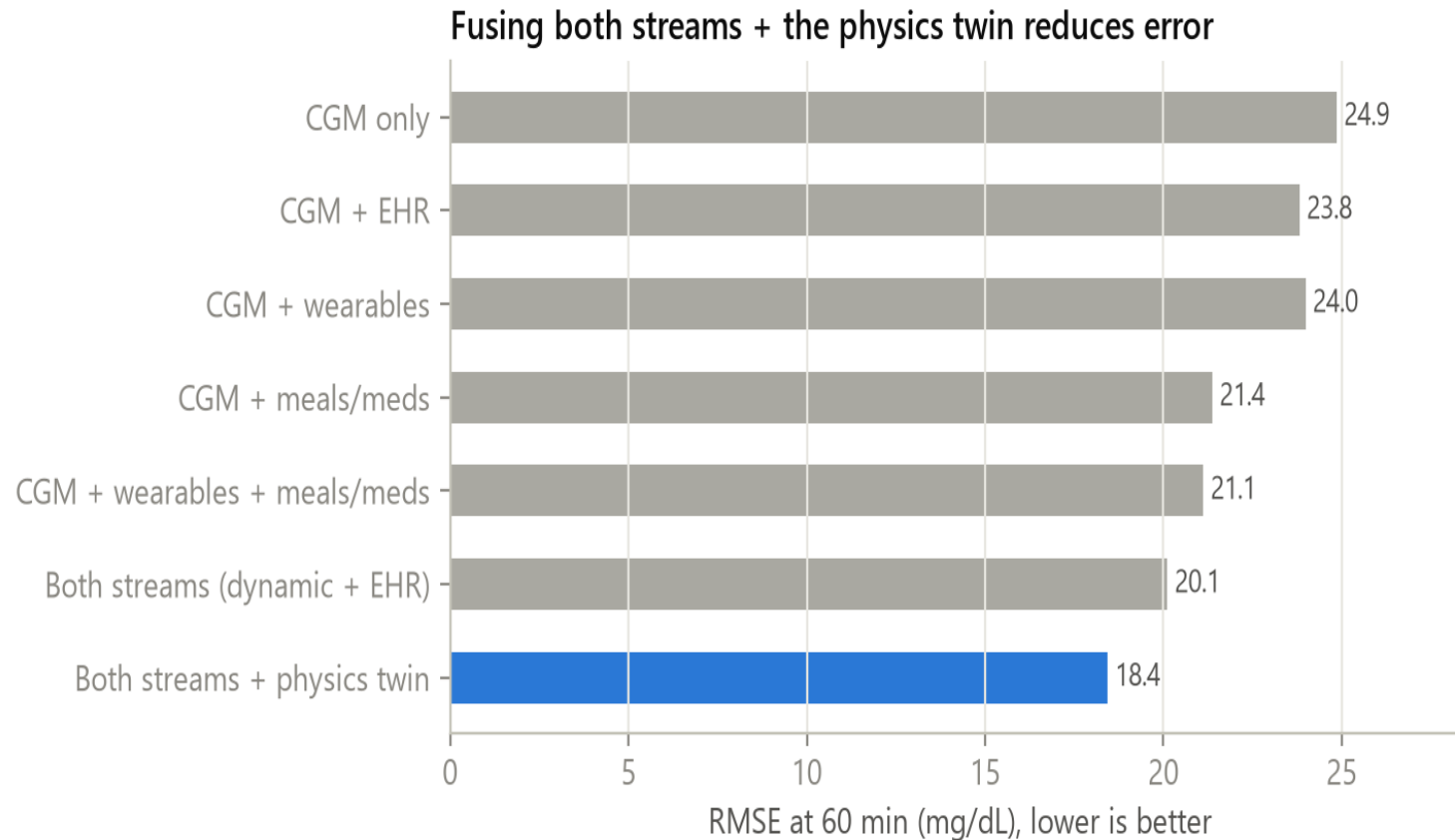


Decision curves: spike alerts give net benefit across thresholds



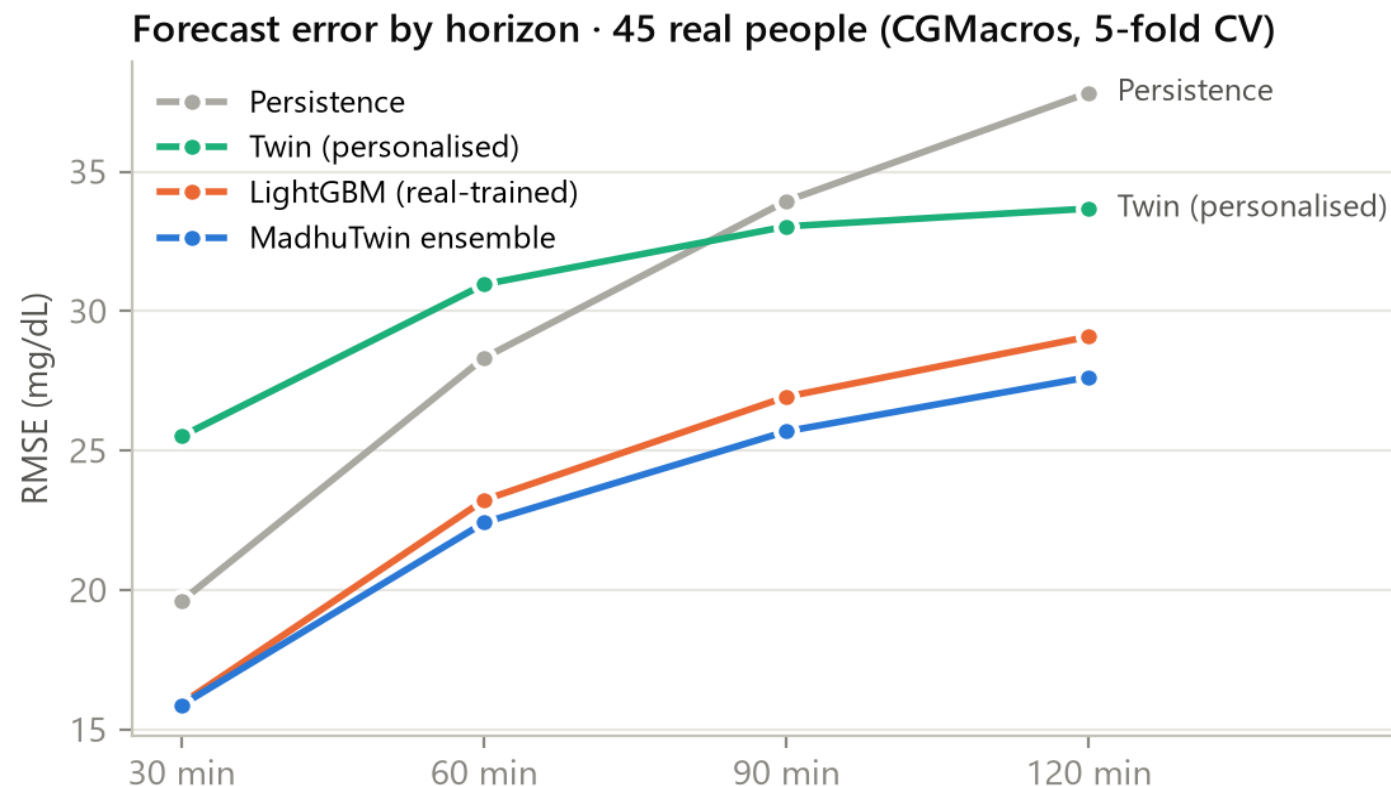
- **Calibrated:** after recalibration on held-out patients (class-weighted training inflates raw risks), a 30% spike risk comes true about 30% of the time
- **Useful:** alerting on MadhuTwin's risk gives more net benefit than alerting everyone or no one across thresholds
- **Fair:** 1-hour error ranges 12.8–22.7 mg/dL across 19 subgroups of sex, age, BMI (Asian cut-offs), status, therapy and region

Fusing both streams matters; the twin adds most



- CGM alone: 24.9 mg/dL at 60 min; spike AUROC 0.92
- Both streams (wearables + meals/meds + EHR): 20.1; AUROC 0.95
- + physics twin: 18.4; AUROC 0.96
- Each stream helps; the EHR helps most in combination, telling the model whose body the wearables describe

It transfers to real people and to an external population



60-min RMSE (mg/dL)	CGMacros	BIG IDEAs	Shanghai
Persistence	28.3	23.0	27.5
TwinNet, synthetic only	26.4	20.7	32.9
TwinNet, sim-to-real	23.5	19.6	22.3
MadhuTwin ensemble	22.4	19.0	21.1
Clarke A+B, MadhuTwin	99.5%	99.3%	98.9%

45 CGMacros and 16 BIG IDEAs participants (real wristband heart rate and HRV), 5-fold patient cross-validation, and 109 ShanghaiT2DM recordings from a different country, sensor and diet, with no wearables at all.

CGM-light: one week of CGM, then fingersticks

An affordability mode for India: the twin keeps estimating continuous glucose after the sensor comes off

6.4%

twin MARD vs true glucose (carry-forward fingersticks: 23.8%)

95%

of twin estimates within 20% of true glucose (carry-forward: 56%)

± 3.8 pp

error in weekly time-in-range (fingersticks alone: ± 5.4 pp)

How it works

Week 1 personalises the twin on CGM. In week 2 only 4 fingerstick checks a day remain; the Unscented Kalman Filter fuses them with logged meals, doses and smartwatch activity to estimate glucose in between. Evaluated on 199 held-out synthetic patients against their true glucose; because the simulator shares the twin's structure this is an upper bound until tested on real CGM-off data.

Robust, fast and honest about its limits

Three questions any deployment raises

1 · No smartwatch, no records?

Missing at inference	RMSE 60	Spike AUROC
All data	19.0	0.96
No smartwatch (wearables missing)	19.5	0.95
No EHR	19.4	0.95
No meal or dose logs	19.1	0.95
No physics twin	21.2	0.95
Only CGM (no smartwatch, EHR, logs or twin)	27.3	0.92
50% of past CGM readings lost	19.3	0.95

One deployed model; data removed at inference (synthetic test patients). The physics twin is the input it misses most.

2 · Can the twin be trusted for a whole day?

7.0 mg/dL **2.8 pp**

median error in daily mean
glucose, 24-h replay (CGMacros)

median time-in-range error
(ShanghaiT2DM: 7.3 pp)

Short lows are not reproduced over 24 h, so hypo warnings come from the 2-hour forecast; each patient's replay is shown as a trust check.

3 · What does it cost to run?

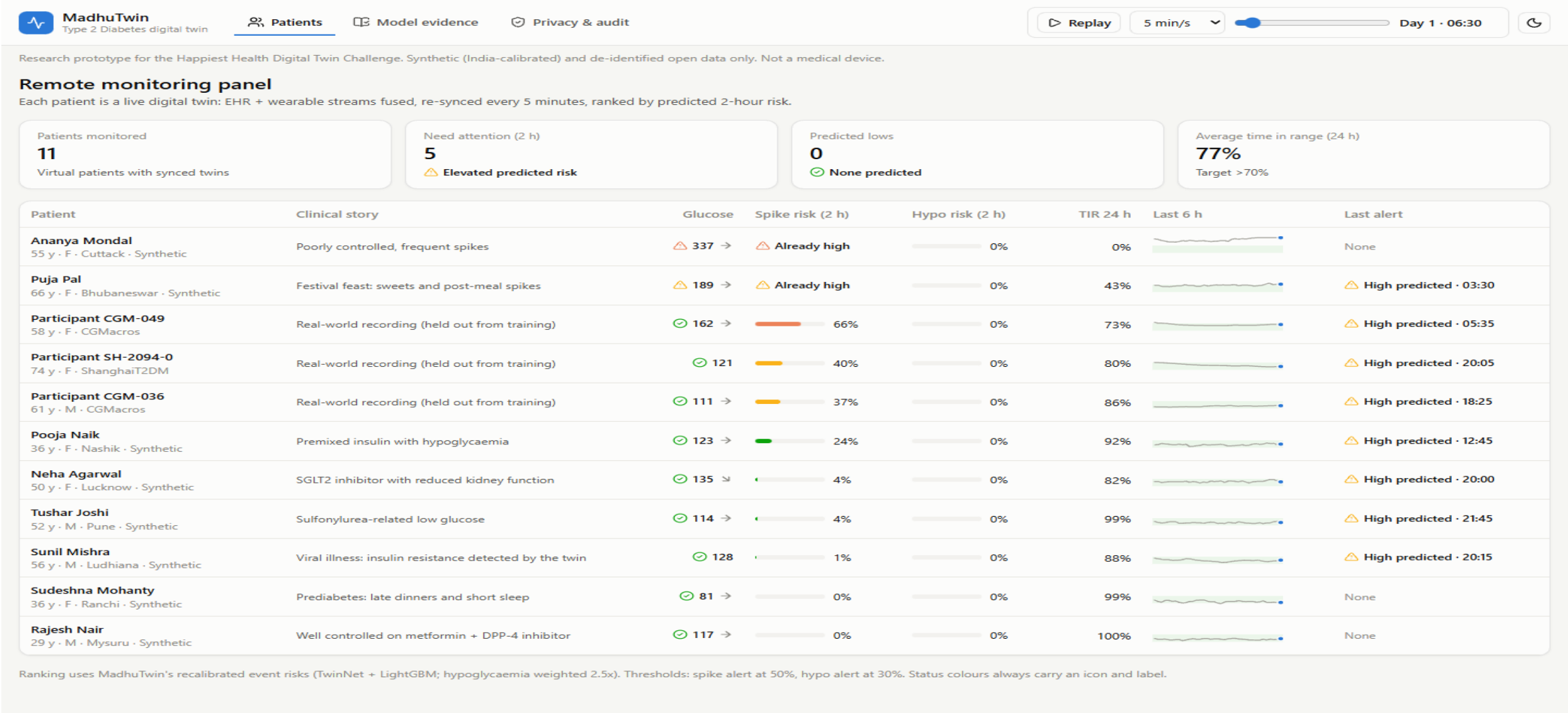
2.2 ms

per 2-hour forecast on a laptop
CPU, no GPU

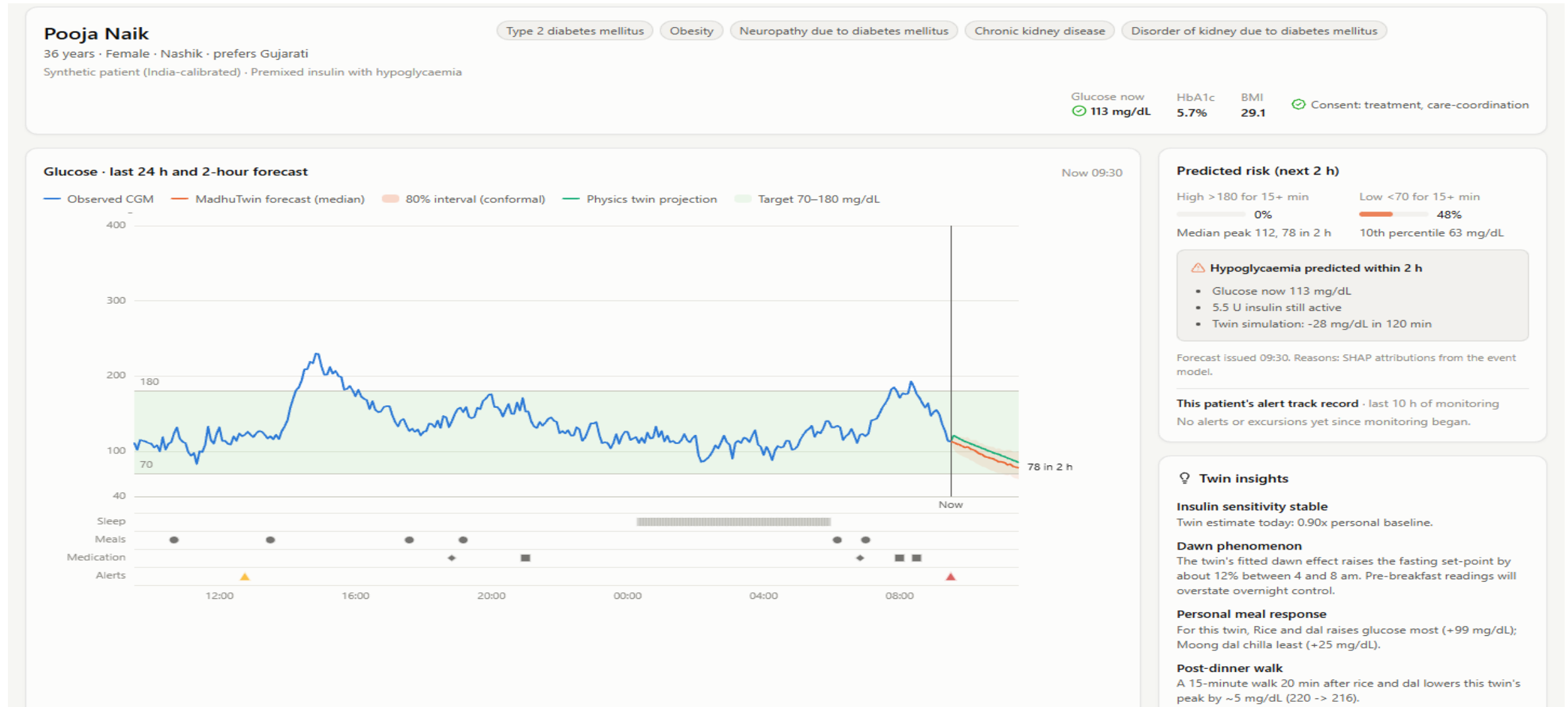
1.8 s

to personalise a new patient (twin +
TwinNet)

The doctor's view: a risk-ranked panel

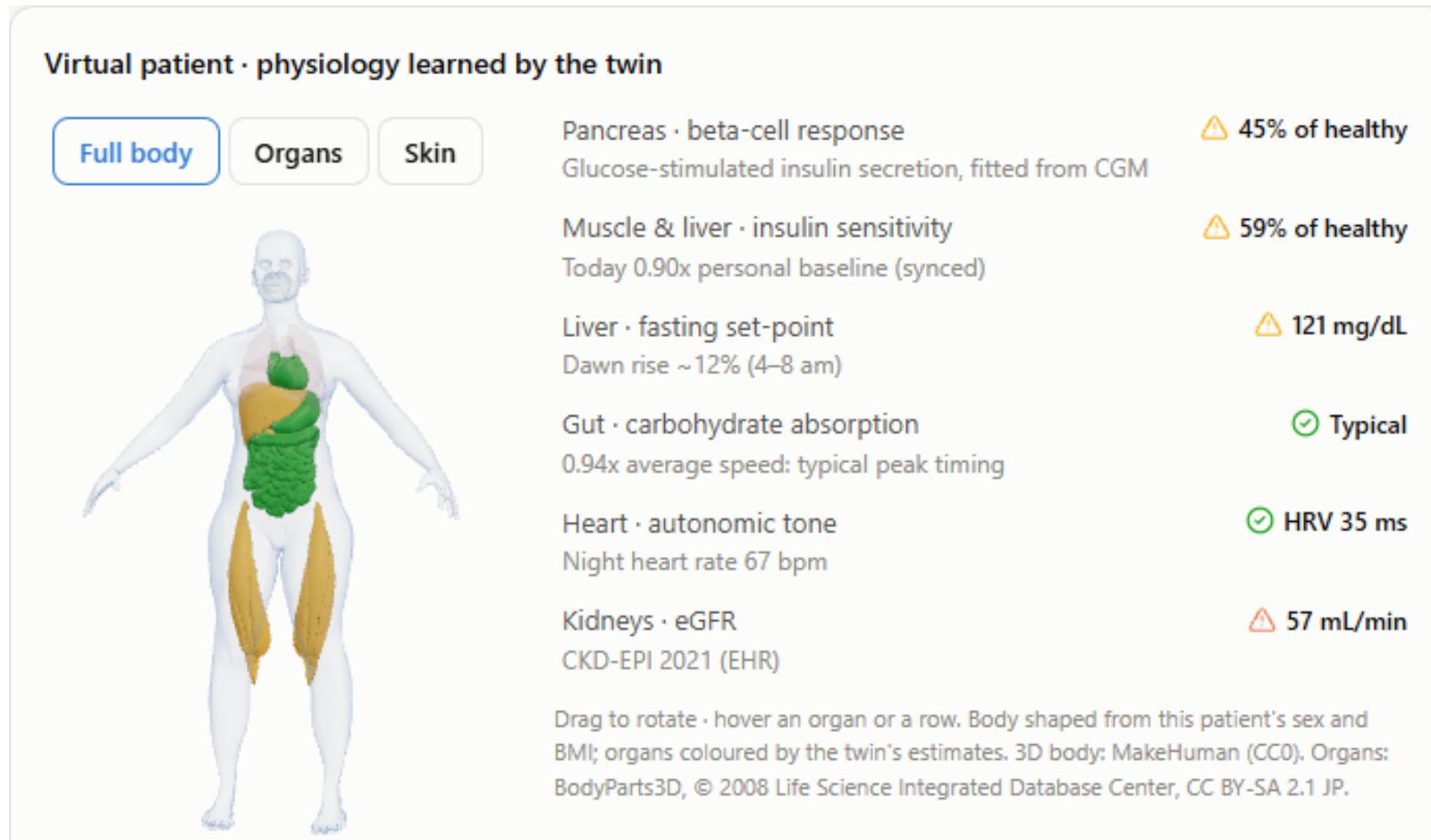


Virtual patient: forecast, reasons, physiology



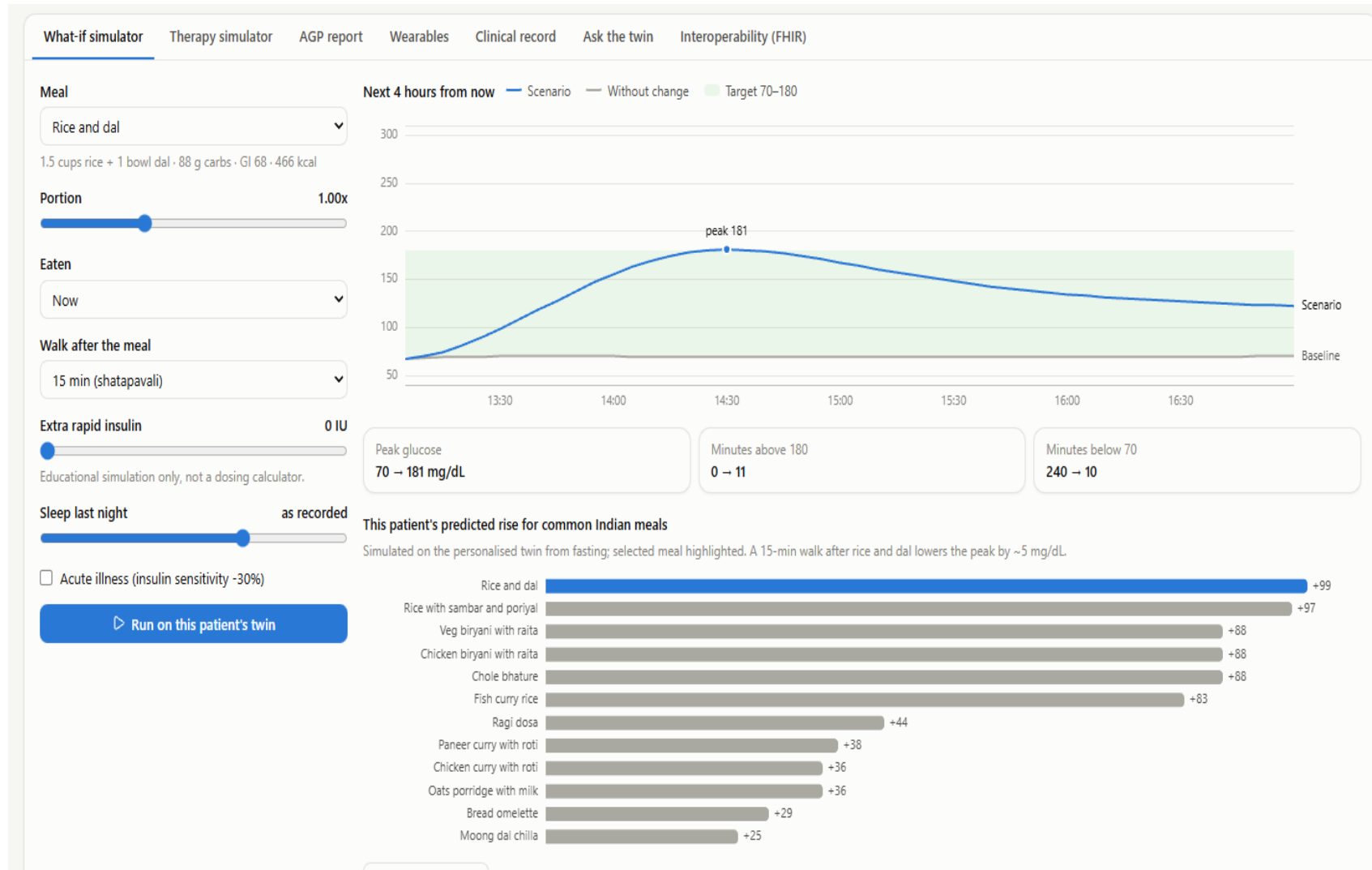
The virtual patient, in 3D

A real human body shaped like the patient, with anatomical organs coloured by the twin



- Body from the MakeHuman mesh (CC0), shaped from the patient's sex and BMI in the EHR
- Anatomical organs from BodyParts3D, placed by skeletal landmarks (2 cm mean fit error)
- Each organ coloured by the twin's estimate: beta-cell response, insulin sensitivity, fasting set-point, absorption, kidney function
- The heart beats at the patient's own night heart rate; rotate, zoom, hover an organ

What-if on their twin, nudges in their language



Try a prescription on the twin first

Neha Agarwal: SGLT2 inhibitor with eGFR 51. Should it be stopped?



86.5% → 72.6%

time in range over the next 24 h if the SGLT2 inhibitor is stopped

88.5%

keeping it and adding a DPP-4 inhibitor: as much as doubling Gliclazide (88.5%)

11.4 mg/dL

trust check: the twin's replay of the last 24 h vs CGM (agreement good)

Responsible by design

1

DPDP Act 2023

Consent artefacts, purpose limitation, data minimisation, purpose-tagged audit trail; synthetic and de-identified data only

2

ABDM-ready interoperability

FHIR R4 with SNOMED CT, LOINC, WHO ATC; CGM as SampledData; the forecast exported as a RiskAssessment

3

Explainable

Reasons on every alert (SHAP), a visible trust-in-physics gate, organ-level physiology, AGP report

4

Honest evaluation

Patient-level splits, external validation, bootstrap CIs, model card with limitations

5

Regulatory path

Decision support, not autonomous treatment: CDSCO Software as a Medical Device route with prospective validation

From proof of concept to Indian clinics



Now

Proof of concept

Twin, TwinNet, dashboard, FHIR;
validated on synthetic, US and Chinese
data



3 months

Co-design

Diabetologist review of alerts and
thresholds; native-speaker review of
patient text



6 months

Pilot

50–100 people with T2D in Bengaluru
clinics: CGM + smartwatch + ABDM-
linked EHR



12 months

Validation

Prospective accuracy and outcome
study; CDSCO SaMD submission

Impact we are aiming for

Fewer nocturnal hypoglycaemia episodes in people on sulfonylureas and premixed insulin, more time in range through personalised meal and activity advice, and a care team that sees risk before it becomes an emergency.

Thank you

MadhuTwin: model · sync · simulate, for every person living with diabetes

Live demo: madhutwin.onrender.com · Code: github.com/ShubAce/MadhuTwin · Apache-2.0

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